

Mechanics of Vibrations

Module 1 - Stress Doc

Module 1: Vibrations

Stress Concept -- Sample text with symbols: A B, P Q, damping and , em--dash, caf

A concept block carrying unicode: Sample text with symbols: A B, P Q, damping and , em--dash, caf and inline math $\forall x \in \mathbb{R}, \exists y : y > x$.

$$\int_a^b f(x) \, dx = F(b) - F(a)$$
 and chemistry
$$2 \text{H}_2 + \text{O}_2 \longrightarrow 2 \text{H}_2\text{O}$$

Appears in every mechanics/vibrations exam.

SHM - damping

Damped Harmonic Oscillator

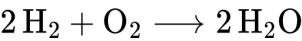
$$\frac{d^2y}{dx^2} + \omega^2 y = 0$$

Symbol	Meaning	Unit
ω	angular frequency	rad/s
ζ	damping ratio (unicode)	-

Example

A Sample text with symbols: A B, P Q, damping and , em--dash, caf. Given the table below: | Trial | Load (N) | Deflection (mm) | Stress (MPa) | Remarks | |---|---|---|---|---| | 1 | 100 | 0.42 | 12.5 | within limit | | 2 | 200 | 0.88 | 25.1 | within limit | | 3 | 300 | 1.31 | 37.8 | within limit | | 4 | 400 | 1.79 | 50.2 | near yield | | 5 | 500 | 2.35 | 62.9 | yield onset | | 6 | 600 | 3.02 | 75.4 | plastic | | 7 | 700 | 3.91 | 88.0 | plastic | | 8 | 800 | 5.20 | 100.6 | necking | | 9 | 900 | 7.10 | 113.1 | necking | | 10 | 1000 | 9.85 | 125.7 | fracture | Find deflection at 500N using $\frac{d^2y}{dx^2} + \omega^2 y = 0$.

Applying $\int_a^b f(x) \, dx = F(b) - F(a)$ and the boundary condition $\forall x \in \mathbb{R}, \exists y : y > x$, and the reaction



, we solve the system $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ to get the deflection.

Applies when:

Valid when $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is non-singular.

Damping Regimes

	Condition	Formula	Behaviour
Underdamped	$\zeta < 1$	$\frac{d^2y}{dx^2} + \omega^2y = 0$	Sample text with symbols: A B, P Q, damping and , em--dash, caf
Critically damped	$\zeta = 1$	$x = Ae^{-\omega t}$	fastest return
Overdamped	$\zeta > 1$	$x = Ae^{r_1t} + Be^{r_2t}$	no oscillation